

Testing & QA Specification

Student OS Documentation — File 14

Testing Levels

Unit tests for utilities/domain logic; integration tests for API/database flows; component tests for important UI behavior; end-to-end tests for critical journeys; manual responsive/accessibility checks.

Critical Journeys

Login/logout

Create and complete task

View Today

Create timetable entry

View calendar

View attendance

Create exam

Create/edit note

Receive/read notification

Change settings

Validation Tests

Required fields, invalid dates, duplicate/overlap rules, unauthorized IDs, empty datasets and large text/input values.

Security Tests

Protected routes, ownership checks, token/session handling, rate limits, injection resistance, upload validation and secret exposure.

UI Tests

Desktop/tablet/mobile layouts, dark mode, loading, empty, error, success, keyboard navigation and focus states.

Charts Tests

Correct calculations, zero-data behavior, responsive rendering, labels/tooltips and date filters.

Performance Tests

API response behavior for realistic datasets, dashboard load, large task lists, calendar ranges and chart rendering.

Release Gate

No critical test failures; build/lint passes; major user flows verified; security-sensitive paths reviewed; responsive smoke test completed.